

Vigno Smart Class — License System

Document 1 of 2 · High-Level Design (HLD)

Applying the Xbox / PlayStation "account-owns-a-license" model to a MERN web platform

Companion: Document 2 — Detailed Technology & Implementation Plan

1. The Core Principle

Same truth that powers Xbox and PlayStation: you don't buy the game's files, you buy a **license** — a signed permission that says "this account may use this content."

License = a signed permission. The content is verified through the license + the account, never by trusting the file sitting on a device.

On Xbox/PlayStation a purchase is recorded against your Microsoft / PlayStation account; the console downloads the files but checks the **license** before letting the game run. Vigno copies this exactly: a **License Authority** on our server issues a cryptographically signed license after payment, and every access re-checks it.

2. What Xbox & PlayStation Do (Observed Model)

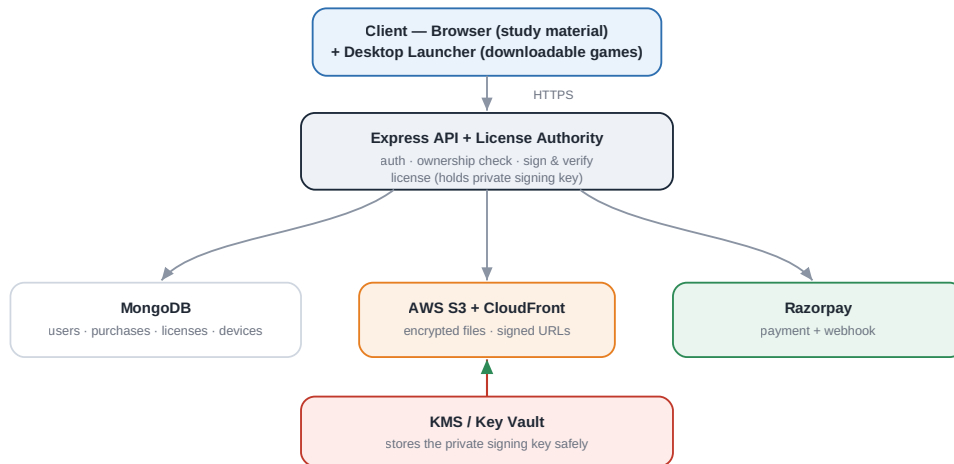
Confirmed from xbox.com and playstation.com plus how these platforms behave.

Idea	Xbox	PlayStation	Vigno equivalent
Identity	Microsoft account	PlayStation Network account	Vigno account (JWT login)
Buy = license, not files	"Software Usage Terms"	"Software Usage Terms"	Signed license record
Outright vs subscription	Buy game OR Game Pass	Buy game OR PS Plus	Buy content OR (future) plan
Online needed at key moments	Account + activation	"Account + internet connection required"	Login + license issue/verify
Re-download anytime	From your Library	From your Library	From "My Purchases"
Codes / gifting	"Redeem Code"	PSN codes	Redeem / coupon (optional)

Marketing pages don't reveal internals, but both clearly run the same pattern: **purchase tied to an account → license issued → files delivered → license verified before use.**

3. High-Level Architecture

The new piece compared to a plain store is the **License Authority** — the trusted signer that mints and verifies licenses.



The **private signing key** never leaves the server/KMS. The matching **public key** is the only thing shipped to clients, and it can only *verify* — never create — a license.

4. What a License Contains

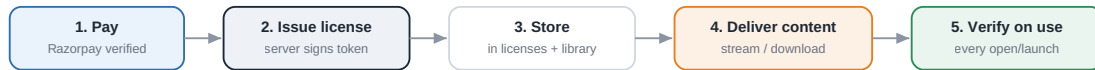
A small signed packet — like the Xbox license token. Human-readable fields plus a cryptographic signature.

```
License (signed token)
├─ licenseId  unique id
├─ userId    who owns it
├─ contentId  which file/game
├─ type       stream | download
├─ deviceId   (for downloadable software only)
├─ issuedAt / expiresAt
├─ status     active | revoked
└─ signature  ← cryptographic proof from our server
```

The **signature** is the important part. Anyone can read the fields, but only our server (with the private key) can produce a valid signature — and any edit to the fields breaks it.

5. License Lifecycle (Purchase → Use)

From buying to opening, mirroring the Xbox "buy → added to account → install → verify on launch" flow.



6. Where the License Applies (Two Content Lanes)

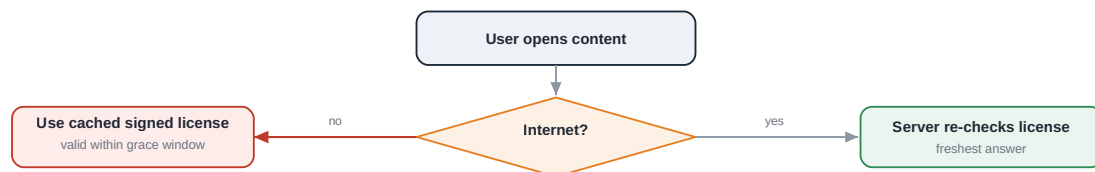
The license rule is the same; only the delivery and verification point differ.

Content	Delivery	License checked...	Closest console analogy
Study material (PDF, video)	Stream only, in-browser	On the server , before each signed URL	Cloud-streamed game (PS Plus streaming)
Software (.exe, game)	Encrypted download	By the launcher , before it decrypts & runs	Installed console game checked at launch

Study material never leaves the browser, so it can't be copied. Downloadable software *can* be copied, so the license check moves into a launcher app — exactly how a console verifies an installed game.

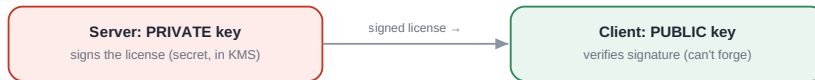
7. Online vs Offline Verification

Consoles allow limited offline play on a "home" device. Vigno's launcher uses the same idea with a short grace period.



8. Trust Model — Why a License Can't Be Faked

Public-key cryptography, the same foundation Microsoft/Sony use. The server signs; clients only verify.



Edit any field → signature breaks → rejected. Attacker has no private key → cannot create a valid new license. (Cracking instead targets the launcher code — see Doc 2 security.)

9. High-Level Security Principles

Principle	Meaning
Account > file	Ownership is the signed license tied to an account, never the bytes on disk.
Sign on server only	Private key stays in KMS; clients get the public key to verify.
Verify every time	Re-check the license before each stream URL or launcher run.
Short-lived everything	Signed URLs (~60s) and licenses with expiry + refresh.
Bind & trace	Downloads bound to a device; PDFs/videos watermarked to trace leaks.
Revocable	A license can be revoked (refund/fraud) and stops working on next check.

10. Component Map

Vigno Platform

- └─ Web App (React) – browse, buy, in-browser viewer
- └─ API Server (Express)
 - └─ Auth & Roles
 - └─ License Authority (sign / verify / revoke)
 - └─ Ownership & signed-URL service
 - └─ Payment + webhook
- └─ Data (MongoDB) – users · purchases · licenses · devices
- └─ Storage (S3 + CloudFront) – encrypted files
- └─ KMS – private signing key
- └─ Desktop Launcher (only for downloadable .exe/games)

Continue to **Document 2** for exact technologies, the signed-token format, database schema, APIs, the step-by-step build plan, and detailed security.